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Predicted the Rate of Extra-Urban Accidents in Ten Iranian Provinces Using Artificial Intelligent based on Influencing Factors

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Abstract— The rising mortality rate associated with intra-urban and extra-urban road accidents constitutes a pressing socio-welfare challenge within the nation. Consequently, this research seeks to forecast accident rates in extra-urban regions across ten provinces in Iran and to pinpoint the primary factors that could facilitate their mitigation. To achieve this objective, Multilayer Perceptron (MLP) neural networks were utilized to assess and determine the most significant contributing factors. Data pertaining to extra-urban accident occurrences, along with essential road enhancement elements, were gathered for the period spanning 2021 to 2023. Upon applying the ANN-MLP model, the findings revealed a prediction accuracy of 85.3% concerning extra-urban accidents. The MLP analysis identified the total number of operational road maintenance stations, the quantity of technical inspection center lines, and the effectiveness of asphalt distribution as the most pivotal factors affecting extra-urban accidents. The insights derived from this study can aid planners and authorities in recognizing high-risk zones and enhancing road infrastructure. Furthermore, these results lay a robust groundwork for formulating new traffic safety policies and executing preventive strategies, such as improving driver education and fostering a culture of safety within the community.

Index Terms-- Artificial Intelligence, Neural Networks, Classification, Extra-Urban, Accident.

I. INTRODUCTION

Urban transportation has increasingly emerged as a critical issue, primarily due to the rapid expansion of the population along with the continuous development of urban infrastructure. This phenomenon has made urban transportation one of the most pressing concerns today, as it directly influences the overall quality of life and presents significant challenges to environmental sustainability. As the number of vehicles on the roads continues to grow, we are witnessing a corresponding rise in various problems, including traffic congestion, vehicular accidents, and

escalating air pollution levels. These issues not only pose immediate threats to the efficiency of urban infrastructure but also place additional strain on the surrounding environment [1]. As a result, the complications arising from urban transportation systems contribute to heightened medical expenses, which in turn create a considerable financial burden on individual households as well as on the healthcare industry at large. To effectively address and mitigate these challenges, it is essential to acquire a comprehensive understanding of the myriad factors that influence road accidents [2]. Additionally, the ability to accurately predict accident conditions and associated mortality rates plays a pivotal role in the broader scope of accident prevention measures. In light of these pressing concerns, many domestic and international studies have been undertaken, each aimed at uncovering the intricate dynamics surrounding road safety. Some of these studies will be briefly reviewed in the following section to shed light on the findings and implications of this vital area of research [1].

Ardehali Zadeh et al. (2018) analyzed the effects of pavement conditions and maintenance on accidents on 19 major roads in Tehran through regression and logarithmic analyses. Their findings revealed that pavement defects significantly affect accidents related to road conditions, underscoring the importance of proper pavement maintenance [1]. Kamboozia et al. (2021) examined accident rates on rural roads in Gilan Province using multiple logistic regression models, MLP neural networks, and radial basis functions. They found that pedestrians experience the highest mortality rates, primarily due to the proximity of villages to roads [2]. Barggol et al. (2022) investigated factors influencing the severity of heavy vehicle accidents on intercity roads in Gilan Province through logistic regression. Their findings showed that road geometry, head-on collisions, non-peak hours (1–6 a.m.), peak evening hours (7 p.m. to midnight), local drivers, and collisions with agricultural machinery contribute to increased accident



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severity [3]. Taheri et al. (2023) examined road safety on intercity routes in Khorasan Razavi Province over four years using accident severity prediction models. Their logistic models identified "heavy vehicle flow" and "speed exceeding 85 km/h" as the most significant factors influencing accidents in the region [4].

Pathik et al. (2022) proposed an intelligent accident detection and rescue system that emulates human cognitive functions using IoT and AI technologies. The study employed the InceptionResNetV2 model to detect accident-related patterns in images. By integrating IoT with real-time data analysis, the system provided immediate alerts to emergency teams, significantly reducing response times. This technology proved effective in both congested urban areas and rural roads, achieving 98% accuracy and high processing speed [5]. Ranyal et al. (2022) investigated how smart sensors and data collection platforms, including smartphones, drones, and vehicles equipped with non-invasive sensors like RGB and thermal cameras, lasers, and GPR sensors, can identify road damage patterns such as cracks and potholes. By employing machine learning algorithms like SVM and K-Means, they achieved 91% accuracy in detecting high-risk areas, significantly reducing road maintenance costs and enhancing safety [6]. Rathee et al. (2023) systematically examined how vehicle sensor data combined with deep learning models can identify road defects. Their research utilized CNN-based models optimized for image processing, achieving 94% accuracy in detecting surface issues such as potholes and cracks, ultimately aiming to reduce accidents related to poor road conditions [7]. Zarza et al. (2023) employed Transformer and large language models for accident prediction. Their model analyzed time-series data and identified risk factors through PCA, achieving 95% accuracy in predicting accident probabilities and enabling proactive safety measures [8].

Based on the comprehensive analysis of various studies, it has become increasingly evident that Iran is positioned among the countries that experience alarmingly high rates of traffic accidents and associated fatalities. This troubling situation can be attributed, in part, to the nation's rapidly growing population coupled with a significant rise in the diversity and number of vehicles on the roads. Road transportation holds the prominent position as the primary mode for the movement of goods and passengers within the country, making the safety and efficiency of these routes extremely critical. A thorough examination of the various factors that influence road safety is essential, alongside efforts to improve road quality, particularly in provinces that are either densely populated or that attract a large number of tourists. Such measures are likely to result in not only enhanced road conditions but also a substantial decrease in

both accident occurrences and mortality rates associated with road incidents.

To address these pressing issues, advancements in technology, particularly in the fields of artificial intelligence (AI) and machine learning (ML), have emerged as cutting-edge solutions that can significantly improve traffic flow, enhance overall road safety, and ultimately reduce the frequency of accidents. The scope of this research is focused explicitly on road accidents, delving into innovative AI and ML-based methodologies aimed at predicting and managing intercity accident scenarios effectively. Data utilized for this in-depth study were meticulously gathered from the statistical yearbooks of the Road Maintenance Organization. This dataset covers a range of relevant metrics that relate directly to road networks and their underlying infrastructure over a comprehensive three-year period, specifically from 2021 to 2023, across ten of the major provinces in Iran. In the course of the analysis, the multilayer perceptron (MLP) neural network model was strategically employed to estimate pivotal indicators that are essential for the enhancement of road conditions. The study rigorously constructed and analyzed a comprehensive table of predictive variables, accompanied by a detailed neural network diagram, to illustrate the intricate relationships between various factors influencing road safety. The insights gained from this research hold significant potential to contribute to a meaningful reduction in intercity accident rates, while also fostering a safer environment on urban roads throughout Iran. By leveraging AI and ML technologies, we aim to pave the way for smarter and more reliable road safety solutions that can ultimately save lives and improve travel experiences for all road users.

II. METHODOLOGY

An Artificial Neural Network (ANN) is a sophisticated computational model that draws inspiration from biological systems, particularly the way neurons function in the human brain. At its core, an ANN comprises a collection of interconnected units known as neurons [9]. Each individual neuron plays a critical role by assigning specific weights to the various input signals it receives. It then calculates a weighted sum of these inputs and subsequently processes this sum by applying an activation function, which serves to introduce nonlinearity into the model. This fundamental process allows the network to effectively learn and recognize complex nonlinear patterns inherent in the data it is trained on [9].

Neural networks come in different forms, each tailored for specific tasks, including single-layer perceptrons and multilayer perceptrons (MLP). The current study focuses specifically on the multilayer perceptron architecture.



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Typically, an MLP is structured around three primary layers: the input layer, which is responsible for receiving and processing raw data; the hidden layer, which carries out intricate computations on the inputs and subsequently transmits the resulting outputs to the following layer; and finally, the output layer, which plays a pivotal role in determining the final output of the network after all processing is complete. The learning methodology employed by an MLP is characterized as feedforward; in this mode, data flows from the input layer through the hidden layers to reach the output layer, with the output being computed at the end of this pathway [9]. This straightforward yet effective approach supports the network in refining its performance based on the input it receives and the corresponding outputs it generates. The mathematical representation of an MLP is expressed in the following form:

$$Z = \sum_{i=1}^n x_i w_i + b \quad (1)$$

Where $x_i, (i=1, \dots, n)$ are the inputs, $w_i, (i=1, \dots, n)$, b is the bias value and Z is the output of the network. The MLP model is specifically designed to tackle nonlinear problems effectively, which necessitates the use of activation functions in both the hidden layers and the output layer. These activation functions, often referred to as "transfer functions," play a crucial role in the overall functioning of the model. Initially, they combine the input values that a particular node receives in order to generate an output. This output is subsequently fed into the activation function of the neural network, where the true transformation occurs. The activation function is essential as it takes the linear combination of inputs and converts it into a nonlinear output. This transformation is critical because it allows the model to capture complex patterns within the data. Furthermore, the activation function also maps the input values to a specific range, which varies based on the type of activation function applied. In this particular study, we have chosen to utilize the Tanh activation function for the hidden layer. This function is known for its capability to produce values within a range of -1 to 1, making it suitable for many scenarios. On the other hand, in the output layer, we have opted for the Identity activation function. This particular function is defined in such a way that it allows the output to remain unchanged, essentially passing through the value as is. The following equations delineate the precise definitions and behaviors of these chosen activation functions, underscoring their significance in the structure and efficacy of the MLP model.

$$\text{Tanh} : f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (2)$$

$$\text{Identity} : f(x) = x \quad (3)$$

III. CASE STUDY

The statistical population involved in this study comprises a total of 10 significant provinces situated of Iran. These provinces have been meticulously chosen due to their distinctive geographical features, economic circumstances, or notable traffic conditions, all of which contribute to the diversity of the sample. The provinces selected for this analysis include Tehran, which is the capital and a major metropolitan area, Khorasan, known for its historical significance, Isfahan, famous for its cultural heritage, Fars, which has a rich history, East Azerbaijan, characterized by its vibrant economy, Mazandaran, recognized for its natural beauty, Qom, a center of religious importance, Gilan, known for its green landscapes, Golestan, with its diverse ecosystems, and Alborz, which features striking mountainous regions. The data used for this comprehensive study have been meticulously gathered from the statistical yearbooks published by the Iranian Road Maintenance and Transportation Organization. These yearbooks provide an extensive range of relevant data and cover the period from the year 2021 to 2023, allowing for a thorough analysis of the transportation and road maintenance conditions across the selected provinces during these years.

The characteristics and indicators influencing the safety of intercity roads, which were used as inputs for the MLP model, include:

- **T1:** Performance of asphalt distribution on national roads,
- **T2:** Completion of directional and warning signs, curve indicators, and speed limit signs,
- **T3:** Installation and repair of guardrails,
- **T4:** Safety enhancements for guardrail ends, fixed obstacles, bridge transitions, and nose installations,
- **T7:** Operational lighting projects based on road types,
- **T10:** Number of active centers for mechanized inspections of heavy vehicles,



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- **T12:** Total number of active road maintenance facilities,
- **T13:** Number of traffic police stations.

The main goal of implementing the MLP model with the specified features is to predict and analyze the mortality rate from intercity traffic accidents (F1) and assess how enhancing various indicators can lower accidents and fatalities. The MLP model was run using SPSS Modeler software, version 18.

IV. ANALYSIS OF RESULTS

Figure 1 shows the summary of the MLP model.

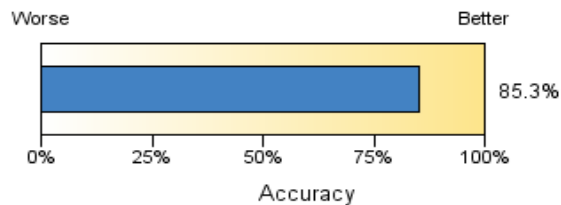


Figure 1. Summary of the MLP Model

The MLP model achieved an accuracy of 85.3%, indicating strong performance in predicting the target index (Figure 1). Figure 2 shows the predictor importance chart, highlighting the significance of each input variable (T1 to T13) for predicting the F1 index. Notably, T13, representing the number of traffic police stations, has the highest importance value of 0.2357, making it the most critical factor in reducing accidents. Table 1 shows the weight of each input index.

Table 1. Importance of Predictors

Feature Weight	Indicators	Feature Weight	Indicators	Feature Weight	Indicators
0.1315	T1	0.1511	T11	0.2357	T13
0.0825	T10	0.0917	T8	0.097	T2
0.0352	T4	0.05	T7	0.0821	T12
		0.0246			T3

Figure 3 provides a detailed illustration of the structure of the MLP neural network. In examining the figure closely, one can observe that the input indicators, which are linked to the hidden layer by darker arrows, possess significantly higher weights than those of the other indicators represented. This difference in weights suggests that these particular input indicators play a more critical role in influencing the output of the network compared to their counterparts that are connected by lighter arrows.

V. CONCLUSION

The factors that contribute to intercity accidents can be ranked in order of their significance, and they are as follows: At the top of the list is the number of traffic police stations spread across the nation, which carries an importance value of 0.2357, indicating that it has the most substantial impact on the occurrence of these accidents. Following closely behind is the total quantity of road maintenance machinery and equipment available throughout the country, which has an importance value of 0.1511. This is accompanied by the performance of asphalt distribution on the nation's roads, which holds an importance value of 0.1315. Together, these factors play a crucial role in enhancing road safety and reducing the likelihood of accidents.

Moreover, there are several additional factors that also contribute to the prediction of intercity accidents, albeit with a moderate level of importance. These include the total number of active maintenance centers that are operational on the country's roadways, the number of active inspection lanes designated for heavy vehicle inspections, and the number of high-risk points that have been successfully removed from the roads. Additionally, the completion of direction and hazard signage, as well as the installation of speed limit signs, also plays a part in this moderate impact, with their importance values falling within the range of 0.0821 to 0.097.

In contrast, there are certain factors that exhibit a lower level of importance when it comes to predicting intercity accidents. These factors include the installation of various types of barriers and repairs, such as guardrails and New Jersey barriers, as well as safety measures implemented at the ends of guardrails. Other factors that fall into this category include rigid obstacles, noses, and the installation of bridge transition zones. Furthermore, the ongoing lighting projects that are being carried out, which are based on the specific type of road, also have the least influence, with their corresponding importance values being 0.05, 0.0352, and 0.0246, respectively. Each of these elements, while still relevant, plays a much lesser role in the overall prediction of intercity accidents compared to the more significant factors mentioned earlier.



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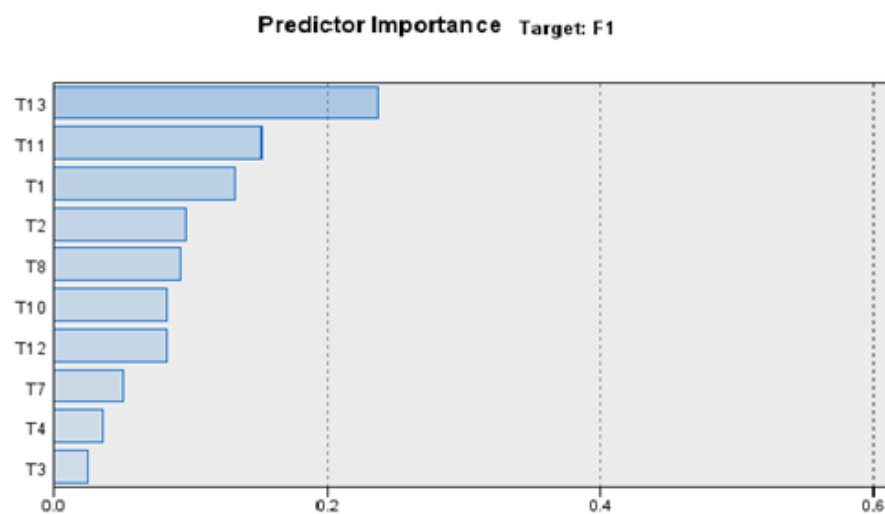


Figure 2. Predictor Importance Chart



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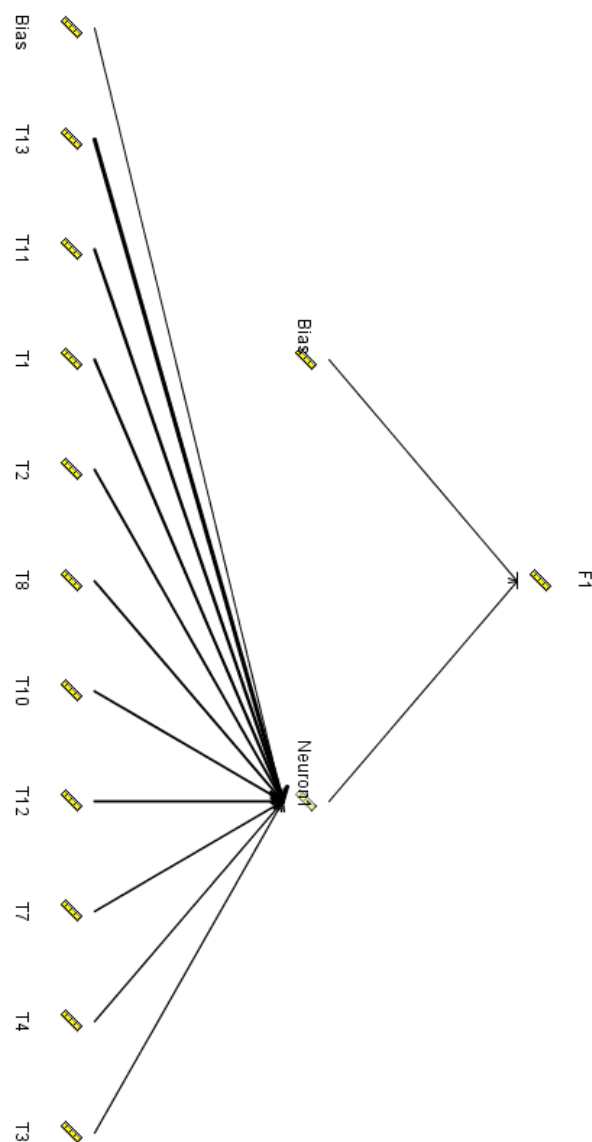


Figure 3. The structure of the neural network for the most important indicators of intercity accidents.

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